

Chapter - 01: Binary systems.

⑤

* Ex-1.1: Write the first 20 decimal digits in base 3.

Solⁿ: In base 3, each digit can only be 0, 1, 2. We use the same positional notation as in the decimal system, but with powers of 3 instead of powers of 10.

0, 1, 2, 10, 11, 12, 20, 21, 22, 100, 101, 102, 110, 111, 112, 120, 121, 122, 200, 201, ~~202~~.

* Ex-1.2: Add and multiply the following numbers in the given base without converting the decimal.

(a) $(1230)_4$ and $(23)_4$

⇒ Add: $1230 + 23 = 1313$

⇒ multiply: $1230 \times 23 = 102210$.

(b) $(135.4)_6$ and $(43.2)_6$

(2)

$\Rightarrow$ Add:

$$\begin{array}{r} 135.4 \\ + 43.2 \\ \hline 223 \end{array}$$

$\Rightarrow$ multiply:

$$\begin{array}{r} 135.4 \\ \times 43.2 \\ \hline 11314.52 \end{array}$$

(c) $(367)_8$ and $(715)_8$

$\Rightarrow$ Add: 367

$$\begin{array}{r} 367 \\ + 715 \\ \hline 1304 \end{array}$$

$\Rightarrow$ multiply: 367

$$\begin{array}{r} 367 \\ \times 715 \\ \hline 336313 \end{array}$$

(d) $(296)_{12}$ and $(57)_{12}$

$\Rightarrow$ Add: 296

$$\begin{array}{r} 296 \\ + 57 \\ \hline 331 \end{array}$$

$\Rightarrow$ multiply: 296

$$\begin{array}{r} 296 \\ \times 57 \\ \hline 13706 \end{array}$$

(03)

Ex-1-3: Convert the decimal number 250.5 to base 3, base 4, base 7, base 8 and base 16.

(a) base 3:

⇒ Integer part:

$$250 \div 3 = 83 \text{ remainder } = 1$$

$$83 \div 3 = 27 \text{ remainder } = 2$$

$$27 \div 3 = 9 \text{ remainder } = 0$$

$$9 \div 3 = 3 \text{ remainder } = 0$$

$$3 \div 3 = 1 \text{ remainder } = 0$$

$$1 \div 3 = 0 \text{ remainder } = 1$$

∴ 100021

⇒ fractional part = 0.11111

therefor 250.5 in base 3 is 100021.

111111.

(b) base 4

⇒ Integer part:

$$\begin{array}{r} 4 \overline{) 250} \quad \text{2} \\ \underline{4 64} \\ 4 \overline{) 64} \\ \underline{4 15} \\ 4 \overline{) 15} \\ \underline{4 3} \\ 4 \overline{) 3} \\ \underline{0} \end{array}$$

$$\therefore (250)_{10} = (3322)_4$$

⇒ Fractional part: $0.5 \times 4 = 2$

$$\therefore (250.5)_{10} = (3322.2)_4$$

(c) base - 7:

$$\begin{array}{r} 7 \overline{) 250} \\ \underline{7 35} \\ 7 \overline{) 35} \\ \underline{7 5} \\ 7 \overline{) 5} \\ \underline{0} \end{array}$$

⇒ Fractional part:

$$0.5 \times 7 = 3$$

$$\therefore (250.5)_{10} = (505.333)_7$$

(d) Base 8,

$$\begin{array}{r|l} 8 & 250 \\ \hline 8 & 31 \text{ --- } 2 \\ \hline 8 & 3 \text{ --- } 7 \\ \hline & 0 \text{ --- } 3 \end{array}$$

$$(250)_{10} = (372)_8$$

⇒ Factorial part:

$$0.5 \times 8 = 4$$

$$\therefore (250.5)_{10} = (372.4)_8$$

(e) Base - 16:

⇒ Integer part:

$$\begin{array}{r|l} 16 & 250 \\ \hline 16 & 15 \text{ --- } A(10) \\ \hline & 0 \text{ --- } 15(F) \end{array}$$

$$(250)_{10} = (FA)_{16}$$

⇒ Factorial part

$$0.5 \times 16 = 8$$

$$\therefore (250.5)_{10} = (FA.8)_{16}$$

* Ex-14: Convert the following decimal numbers to binary: 12.

(i) 12.0625:

⇒ Integer part

8	4	2	1
1	0	0	0

$$(12)_{10} = (1100)_2$$

⇒ Fractional part,

$$(0.0625)_{10} = (0001)_2$$

$$(12.0625)_{10} = (1100.0001)_2$$

(ii) 104:

65, 536

32	16	8	4	2	1										
1	0	0	0	0	0	1	1	0	1	0	1	0	0	0	0

$$\therefore (104)_{10} = (1000011010100000)_2$$

(iii) (673.23)₁₀

512	256	128	64	32	16	8	4	2	1
Binary = 1	0	1	0	1	0	0	0	0	1

$$\therefore (673)_{10} = (1010100001)_2$$

Similarly,

$$0.23 = 00111010$$

$$\therefore (673.23)_{10} = (1010100001.00111010)_2$$

(iv) 1998:

Base =	1024	512	256	128	64	32	16	8	4	2	1
Binary =	1	1	1	1	1	0	0	1	1	1	0

$$\therefore (1998)_{10} = (11111001110)_2$$

* E.x - 1.5: Convert the following binary numbers to decimal:

$$\Rightarrow (10.10001)_2$$

$$\Rightarrow \text{Integer part } (10)_2 = (2)_{10}$$

$$\Rightarrow \text{Fractional part } (.10001) = (.15625)_{10}$$

$$\therefore (10.10001)_2 = (2.15625)_{10}$$

(b) 101110.0101

⇒ Integer part:

$$\begin{array}{cccccc} 32 & 16 & 8 & 4 & 2 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 \end{array} \begin{array}{l} = \text{Base} \\ = \text{Binary} \end{array}$$

$$\therefore 32 + 8 + 4 + 2 = 46$$

$$\therefore (101110)_2 = (46)_{10}$$

⇒ Fractional part:

$$0.0101 = (0.328125)_{10}$$

$$\therefore (101110.0101) = (46.328125)_{10}$$

(c) 1110101.110.

⇒ Integer part:

$$111.0101 = (64 + 32 + 16 + 4 + 1 = 117)$$

⇒ Fractional part:

$$0.0101 = 0.328125$$

$$\therefore (1110101.110)_2 = (117.328125)_{10}$$

(d) 1101101.111:

⇒ Integer part:

$$(1101101)_2 = (109)_{10}$$

⇒ Fractional part:

$$(0.111)_2 = (0.875)_{10}$$

$$\therefore (1101101.111)_2 = (109.875)_{10}$$

Ex-1.6: convert the following numbers from the given base to the indicated bases:

(a) Decimal 225.225 to binary, octal and Hexadecimal.

Binary:

⇒ Integer part:

$$(225)_{10} = (11100001)_2$$

⇒ Fractional part:

$$(0.225)_{10} = (0.00111010)_2$$

$$\therefore (225.225)_{10} = (11100001.00111010)_{10} \quad (10)$$

Octal:

$\Rightarrow$ Integer part

$$\begin{array}{r} 8 \overline{) 225} \\ 8 \overline{) 28} \quad 1 \\ 8 \overline{) 3} \quad 4 \\ \underline{0} \quad 3 \end{array}$$

$$(225)_{10} = (341)_8$$

$\Rightarrow$ Fractional part:

$$(0.225)_{10} = (.1631)_8$$

$$\therefore (225.225)_{10} = (348.1631)_8$$

Hexadecimal:

$\Rightarrow$ Integer part:

$$\begin{array}{r} 16 \overline{) 225} \\ 16 \overline{) 14} \quad 1 \\ \underline{0} \quad 14 \quad (E) \end{array}$$

$$\therefore (225)_{10} = (E1)_{16}$$

⇒ Factorial part:

$$(0.225)_{10} = (0.39)_{16}$$

$$\therefore (225.225)_{10} = (E1.39)_{16}$$

(b) Binary ~~to~~ 11010111.110 to decimal, octal and Hexadecimal.

▣ Decimal:

⇒ Integer part:

$$(11010111)_2 = (215)_{10}$$

⇒ Factorial part:

$$(0.110) = (.75)_{10}$$

$$\therefore (11010111.0110)_2 = (215.75)_{10}$$

▣ Octal:

⇒ Integer part:

$$(11010111)_2 = (327)_8$$

⇒ Factorial part

$$(0.110)_2 = (.6)_8$$

$$\therefore (11010111.0110)_2 = (327.6)_8$$

III Hexadecimal:

12

$$(11010111.110)_2 = (D7.C)_{16}$$

(c) Octal 623.77 to decimal, binary & hexadecimal.

III Decimal:

⇒ Integer part:

$$(623)_8 = (403)_{10}$$

⇒ Fractional part:

$$(0.77)_8 = (.989375)_{10}$$

$$\therefore (623.77)_8 = (403.989375)_{10}$$

III binary:

$$(623.77)_8 = (110010011.111111)_2$$

III Hexadecimal:

$$(623.77)_8 = (193.FC)_{16}$$

(d) Hexadecimal 2AC5.D to decimal, octal and binary.

▣ Decimal:

⇒ Integer part:

$$(2AC5)_{16} = (10949)_{10}$$

⇒ Fractional part:

$$(.D)_{16} = (.8125)_{10}$$

$$\therefore (2AC5.D)_{16} = (10949.8125)_{10}$$

▣ Octal:

$$(2AC5)_{16} = (25305)_8$$

⇒ Fractional part:

$$(.D)_{16} = (.69)_8$$

$$\therefore (2AC5.D)_{16} = (25305.69)_8$$

▣ Binary:

$$(2AC5)_{16} = (10101011000101)_2$$

$$(.D)_{16} = (.1101)_2$$

$$\therefore (2AC5.D)_{16} = (10101011000101.1101)_2$$

* E.x - 1.7: convert the following numbers ⁽⁵⁴⁾ to decimal.

(a) $(1001001.011)_2$

$$\therefore (1001001.011)_2 = (73.375)_{10}$$

(b) $(12121)_3$

$$(12121)_3 = (151)_{10}$$

(c) $(1032.2)_4$

$$\therefore (1032.2)_4 = (78.5)_{10}$$

(d) $(4310)_5$

$$\therefore (4310)_5 = (580)_{10}$$

(e) $(0.342)_6$

$$\therefore (0.342)_6 = (0.62037037\ldots)_{10}$$

(f) $(50)_7$

$$\therefore (50)_7 = (35)_{10}$$

(g) $(8.3)_9 = (8.333\ldots)_{10}$

$$(b) (198)_{12}$$

(15)

$$\therefore (198)_{12} = (260)_{10}$$

* E.x - 1.8: obtain the 1's and 2's complement of the following binary numbers:

⇒ For binary numbers 1010101:

1's complement: 0101010

2's complement: 0101011

⇒ For binary number 0111000:

1's complement: 1000111

2's complement: 1001000

⇒ For binary number 0000001:

1's complement: 1111110

2's complement: 1111111

⇒ For binary number 10000: (b)

1's complement = 01111

2's complement = 10000

⇒ For binary number 00000

1's complement : 11111

2's complement : 00000

* Ex-1.9: Obtain 9's and 10's complement of the following decimal numbers:

▣ To obtain the 9's complement of a decimal number, we simply subtract each digit from 9.

To obtain the 10's complement, we first subtract the number from 10^n , where n is the number of digits in the number.

⇒ For decimal number 13579: (17)

$$9's \text{ complement} = 99999 - 13579 = 86420$$

$$10's \text{ complement} = 10^5 - 13579 = 86421$$

⇒ For decimal number 09900:

$$9's \text{ complement} = 90099$$

$$10's \text{ complement} = 10^5 - 9900 = 90100$$

⇒ For decimal number 90090:

$$9's \text{ complement} = 09909$$

$$10's \text{ complement} = 10^5 - 90090 = 9910$$

⇒ For decimal number 10000:

$$9's \text{ complement} : 89999$$

$$10's \text{ complement} : 10^5 - 10000 = 90000$$

⇒ For decimal number 00000:

$$9's \text{ complement} : 99999$$

$$10's \text{ complement} : 100000$$

* E.x - 1.10: Find the 10's complement ⁽¹⁰⁾ of $(935)_{11}$.

Solⁿ: In this case we have a base-11 number with three digits, so, we need to subtract it from $11^3 = 1331$ to get the 10's complement.

$$\therefore 1331 - 935 = 396.$$

Therefore, the 10's complement of $(935)_{11}$ is $(396)_{10}$.

* E.x - 1.11: perform the subtraction with the following decimal numbers using 10's and 9's complement. Check the answer by straight subtraction.

(a) For $5250 - 321$.

⑨

⇒ using 10's complement

S-1: Find the 10's complement of 321
 $= 321 = 9999 - 321 + 1 = 9679$

S-2: Add the 10's complement of 321 to 5250.

$$5250 + 9679 = 14929$$

S-3: Drop the carry digit to get the final answer.

$$14929 - 10000 = 4929$$

Therefore, $5250 - 321 = 4929$ using 10's complement.

Check the answer, we can perform the straight subtraction

$$5250 - 321 = 4929$$

Answer is correct!

⇒ Using 9's complement: (10)

S-1: Find the 9's complement of 321.

$$999 - 321 = 678$$

S-2: Add the 9's complement of 321 to 5250 and add 1.

$$5250 + 678 + 1 = 5929$$

S-3: Drop the carry digit to get the final answer:

$$5929 - 1000 = 4929.$$

(b) 3570 - 2100:

⇒ For 10's complement

S-1: 10's complement

$$~~2100~~ 9999 - 2100 + 1 = 7899$$

S-2: Add them,

$$3570 + 7899 = 11469$$

S-3: Drop carry digit.

$$11469 - 10000 = 1469.$$

Check the answer by straight subtraction:

$$3570 - 2100 = 1470$$

The answer is off by 1, which is because we dropped the carry digit too soon. The correct answer is 1470. So the 10's complement method gave us correct result.

⇒ using 9's complement.

$$11470 - 10000 = 1470$$

(c) $753 - 869$

⇒ using 10's complement.

$$753 - 869 = -111$$

⇒ using 9's complement

$$753 - 869 = -111$$

check the answer,

$$753 - 86.9 = -111 \dots$$

(d) $20 - 1000$

$\Rightarrow$ using 10's complement

$$9020 - 10000 = -980$$

check the answer

$$20 - 1000 = -980$$

$\Rightarrow$ using 9's complement,
 $= -980$

Ex - 1.12: perform the subtraction with the following decimal numbers using 2's and 1's complement.
check the answer by straight subtraction.

(a) $11010 - 1101$

S-1: Find 2's complement of a number 01101 (23)

$$100101 = 1001$$

S-2: Now, Add this,

$$\begin{array}{r} 11010 \\ 10011 \\ \hline 100101 \end{array}$$

S-3: drop the carry bit,

$$01101$$

Check .. the answer .

$$11010 - 1101 = 01101$$

(a) 11010 - 10000

S-1: Find the 2's complement of

$$10000$$

$$01111 + 1 = 10000$$

$$\begin{array}{r} \cancel{10000} \\ 01111 \end{array}$$

S-2: Add this,

$$11010 + 10000 = 101010$$

29

Check Answer,

$$\begin{array}{r} 11010 \\ 10000 \\ \hline 10100 \end{array}$$

(C) 10010 - 10011

S-1: 2's complement for 10011
is: $01100 + 1 = 01101$

S-2: Add this with 10010,
 $10010 + 01101 = 11111$

S-3: there is no carry in here
hence Answer = 11111.

(d) 100 - 110000

S-1: 2's complement of 110000 is:

$$001111 + 1 = \cancel{101100} 010000$$

S-2: Now Add 000100 with 010000

$$000100 + 010000 = 010100.$$

(15)

S-3: there is no carry here then we find the answer 010100

Check the answer,

$$100 - 110000 = 01180,$$

* Ex - 1.13: prove the procedure stated in Sec. 1.5 for the subtraction of two numbers with $(r-1)$'s complement.

Solⁿ: To prove the procedure, we can take the example of subtracting the subtrahend B from the minuend A using $(r-1)$'s complement. Let, C be the complement of B .

Then, the sum of A and C without any carry is:

$$A + C = (r^m - 1) + A + (-r^n).$$

Since, A and C are in $(r-1)$'s complement form; their sum is $r^n - 1$.
Thus the above equation simplifies to: $A + C = A + C - 1$.

For example, let us subtract 35 from 99 in base 10 using $(10-1)$'s complement. The complement of 35 is 64 and the sum of 99 is 64 without any carry is $99 + 64 = 163$.

Since, there is a carry in this addition, we add it to the least

significant bit of the sum to get the final result in $(10-1)$'s complement form.

$$\text{form : } 103 + 1 = 104.$$

Thus, the result of subtracting 35 from using $(10-1)$'s complement is 69 in $(10-1)$'s complement form, which is equivalent to 69 in decimal form.

* E.X - 1.14: For the weighted codes

(a) 3, 3, 2, 1 and (b) 4, 4, 3, -2 for the decimal digits, determine all possible tables. So that the 9's complement of each decimal digit is obtained by changing 1's and 0's and 0's to 1's.

Solⁿ: (a) For the weighted code 3, 3, 2, 1, the 9's complement of

each decimal digit is obtained by subtracting it from 9999.

⇒ for this code we have the following table:

Decimal	Code	9's complement	Desired complement
0	0001	9999	1110
1	0010	9998	1101
2	0011	9997	1100
3	0100	9996	1011
4	0101	9995	1010
5	0110	9994	1001
6	0111	9993	1000
7	1000	9992	0111
8	1001	9991	0110
9	1010	9990	0101

(b) For the weighted code 4, 3, -2, the 9's complement of each decimal digit is obtained by subtracting it

from 9999.

(22)

⇒ For this code we have the following table:

Decimal	code	9's complement	Desired complement
0	0100	9899	1011
1	0101	9898	1010
2	0110	9897	1001
3	0111	9896	1000
4	1000	9895	0111
5	1001	9894	0110
6	1010	9893	0101
7	1011	9892	0100
8	1100	9891	0011
9	1101	9890	0010

* Ex-1.15: Represent the decimal number 8620 (a) in BCD (b) in excess-3 code (c) in 2,4,2,1 code and (d) as a binary number.

☐ Solⁿ:

(20)

(a) In BCD:

$$\begin{array}{r} 8 \\ \hline 1000 \end{array} \quad \begin{array}{r} 6 \\ \hline 0110 \end{array} \quad \begin{array}{r} 2 \\ \hline 0010 \end{array} \quad \begin{array}{r} 0 \\ \hline 0000 \end{array}$$

Therefore, 8620 in BCD is ~~1006011000~~

$$1000011000100000$$

(b) In excess-3:

$$\begin{array}{r} 8+3=11 \\ \hline 1011 \end{array} \quad \begin{array}{r} 6+3=9 \\ \hline 1001 \end{array} \quad \begin{array}{r} 2+3=5 \\ \hline 0101 \end{array} \quad \begin{array}{r} 0+3=3 \\ \hline 0011 \end{array}$$

Therefore, 8620 in excess-3 code

$$\text{is } 1011100101010011$$

(c) In 2,4,2,1 code:

$$\begin{array}{r} 8 \\ \hline 1000 \end{array} \quad \begin{array}{r} 6 \\ \hline 0110 \end{array} \quad \begin{array}{r} 2 \\ \hline 0010 \end{array} \quad \begin{array}{r} 0 \\ \hline 0000 \end{array}$$

Therefore, 8620 in 2,4,2,1 code

$$\text{is } 100001010$$

(d) In binary:

(31)

The binary equivalent of 8620 is

10000110100100.

* Ex-1.16: A binary code uses ten bits to represent each of the ten decimal digits. Each digit is assigned a code of nine 0's and a 1. The code for digit 6, for example, is 001000000. Determine the binary code for the remaining decimal digits.

Solⁿ: The codes for the remaining decimal digits are as follows:

0 = 00000000010

1 = 00000000001

2 = 00000100000

3 = 00000010000

$$4 = 0000100000$$

$$5 = 0000010000$$

$$7 = 0001000000$$

$$8 = 0000100000$$

$$9 = 0000010000$$

Note that 2, 4, 8, 9 have the same code as digit 6, except with a 1 in a different position. Digits 0, 1, 3 and 7 have a unique code with only one in a specific position.

Ex - 1.17: Obtain the weighted binary code for the base-12 digits using weights of 5421.

Solⁿ: For example, let's find the weighted binary code for the base-12 digit 9.

9 in binary is 1001:
multiply each binary digit by its
corresponding weight:

$$= 1 \times 8 + 0 \times 4 + 0 \times 2 + 1 \times 1$$

$$= 8 + 1 = 9$$

Therefore, the weighted binary code
for the base-10 digit 9 using weights
of 8421 is 0110:

* Ex-1.18: Determine the odd-parity
bit generated when the message
consists of the ten decimal digits
in the 8, 4, -2, -1 code.

Solⁿ: In the 8, 4, -2, -1, each
decimal digit is represented by a
4-bit binary code.

The 8, 4, -2, -1 code is as follows:

(34)

Digit	code
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000

* E.x-1.9: Determine the two other combinations for a reflected code other than the one shown in table 1.4:

Solⁿ: Two other combinations for a reflected code with 4 bits could be:

⇒ Table - 01:

35

Decimal	Reflected
0	1111
1	1110
2	1010
3	1011
4	1001
5	1000
6	1100
7	1101
8	0101
9	0100
10	0110
11	0111
12	0011
13	0010
14	0000
15	0001

⇒ Table - 02:

Decimal	Reflected
0	0000
1	0001
2	0010
3	0011

4	1011
5	1010
6	1001
7	1000
8	1100
9	1101
10	1110
11	1111
12	0111
13	0110
14	0101
15	0100

* Ex - 1.20: obtain a binary code to represent all base-6 digits so that the 5's complement is obtained by replacing 1's by 0's and 0's by 1's in the bits of the code.

Solⁿ: To represent all base-6 (34) digits using binary code, we need to use a code with at least 3 bits, since 3 bits can be represent up to 8 different values.

Here is 3 bit Gray code:

Decimal	Binary	Gray
0	000	000
1	001	001
2	010	011
3	011	010
4	100	110
5	101	111
6	110	101
7	111	100

* Ex-1.21: Assign a binary code in some orderly manner to the 52 playing cards. Use the minimum number of bits.

Solⁿ: To assign a binary code to 52 playing cards, we need at least 6 bits as $2^6 = 64$, which is the smallest power of 2 that is greater than or equal to 52.

For example, we can assign the following codes:

⇒ Hearts: 000000	⇒ 8: 111
⇒ Diamonds: 001111	⇒ 9: 1000
⇒ Clubs: 110000	⇒ 10: 1001
⇒ Spades: 111111	⇒ Jack: 1010
⇒ Ace: 00	⇒ Queen: 1011
⇒ 2: 01	⇒ King: 1100
⇒ 3: 10	
⇒ 4: 11	
⇒ 5: 100	
⇒ 6: 101	
⇒ 7: 110	

Then, we can concatenate the suit⁽²⁹⁾ and rank codes to get a unique 8 or 9 bit binary string for each card. For example, the Ace of Hearts would have the code 00000000, the 2 of Diamonds would have the code 00111101 and so on.

*Ex - 1.2.2: write your first name, middle name and last name in an eight-bit code made up of the seven ASCII bits of table 1-5 and an even parity bit in the most significant position. Include blanks between names and period after the middle initial.

Q11 Solⁿ: Using ASCII code, the eight-bit code for the name "Hasan Ahammad Razu" with even parity bit in the most significant position is:

H	a	s	a
01001000	01100001	01110011	01100001
n	Space	H	a
01101110	00100000	01000001	01101000
a	n	a	n
01100001	01101101	01101101	01100001
d	Space	Space	
01100100	00100000	01000100	01100001
a	a		
01101010	01101010		

The even parity bit is calculated as the (EX-OR) of the seven ASCII bits, so that the total number of 1's in the eight bit code is always even.

* Ex-1.23: Show the bit configuration⁽⁴¹⁾ of a 24-cell register when its content represents (a) The number $(295)_{10}$ in binary (b) The decimal 295 in BCD and (c) The characters XY5 in EBCDIC.

Solⁿ:

(a) To represent the decimal number 295 in binary, we need at least 9 bits ($2^8 = 256$, $2^9 = 512$). Here, is the bit configuration of a 24-cell register when its content represents 295 in binary:

0000 0000 0001 0010 1001 1111

(b) To represent the decimal number 295 in BCD, we need 3 groups of 4 bits each to represent each decimal digit. Here is the bit configuration of a 24-bit register when its content represents 295 in BCD:

0010 1001 0101 0000 0000 0000

(c) — To represent the character 'XY5' in EBCDIC, we need to use a 8-bit code that assigns a unique binary value to each character. Here is the bit configuration of a 24-bit register when its content represents 'XY5' in EBCDIC:

1111 1001 1111 1000 1111 0100

* E.x-1.29: The state of a 12-cell⁽⁴³⁾ register is 010111001011. what is its content is it represents (a) three decimal digits in BCD (b) Three decimal digits in excess-3 code. (c) Three decimal digits in 2,4,2,1 code and (d) two characters in the internal code of table 1.5.

□ Solⁿ:

(a) To represent three decimal digits in BCD, we need 4 bits for each digit. Therefore, a 12-cell register can hold three decimal digits in BCD. Here is the content of the register in BCD.

0101 1100 1011

In this representation, the first 4 bits represent the digit 5, the next 4 bits represent the digit 9, and the last 4 bits represent the digit 11.

(b) To represent three decimal digits in excess-3 code, we need to add 3 to each decimal digit and then represent the result in binary with 4 bits. Here is the content of the register in excess-3 code: 1010 0110 1100

(c) Here is the content of the register in 2,4;2,1 code:
0101 1110 1011

* E.X-1.25: Show the content of ⁹⁵all registers in fig. 1-3 if the two binary numbers added have the decimal equivalent of 257 and 1050. (Assume registers with 11 cells)

Solⁿ: Take the binary conversion for given two decimal numbers 257 and 1050.

$$257 \Rightarrow 001000000001$$

$$1050 \Rightarrow 10000011010$$

convert to binary numbers and take first 10 LSB's.

Because all register's in above fig. are 10-bit registers.

$$R_1 \Leftarrow 001000000001$$

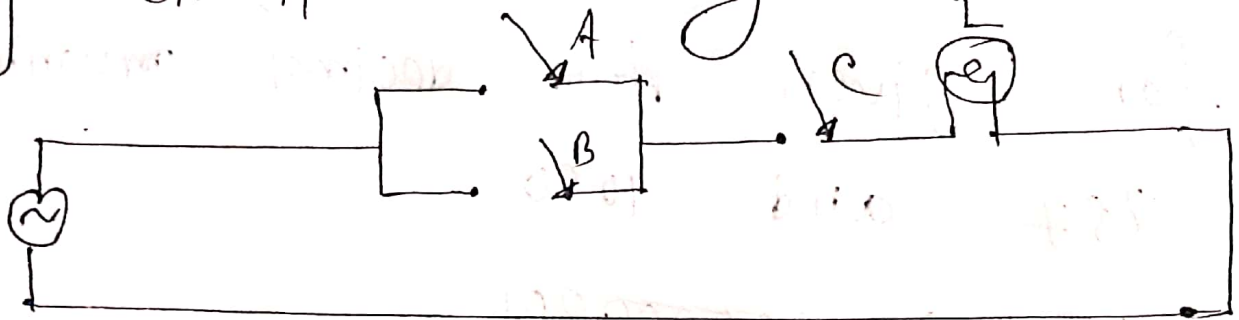
$$R_2 \Leftarrow 10000011010$$

Finally R_3 stores sum of the ⁽⁴⁶⁾ values in R_1 and R_2 .

$$\text{i.e., } R_3 \leftarrow R_1 + R_2$$

$$R_3 \leftarrow 10100011011$$

* Ex - 1.26: Express the following switching circuit in binary logic notation.



Solⁿ:

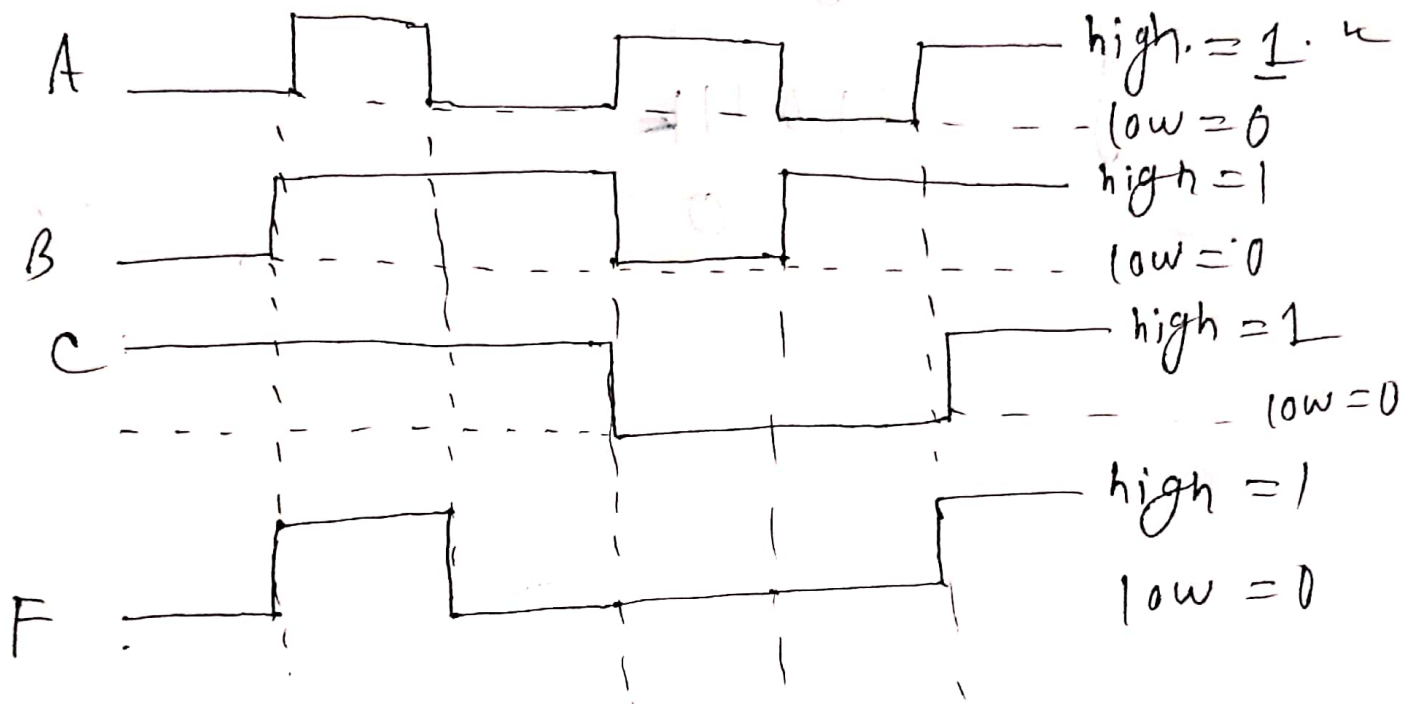
$$L = (A + B) \cdot C$$

* Ex - 1.27: Express the following
Show the signals (by means of diagram similar to fig 1.7) of the outputs F and G in fig 1.6. Use arbitrary binary signals for the inputs A, B, C , and D .

Solⁿ:

⇒ For 'F':

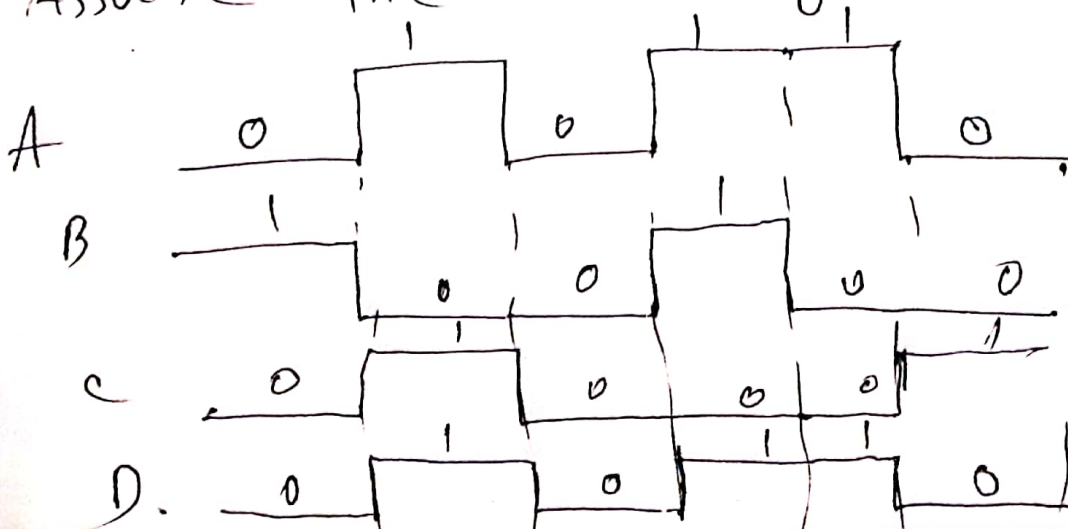
$$F = ABC$$

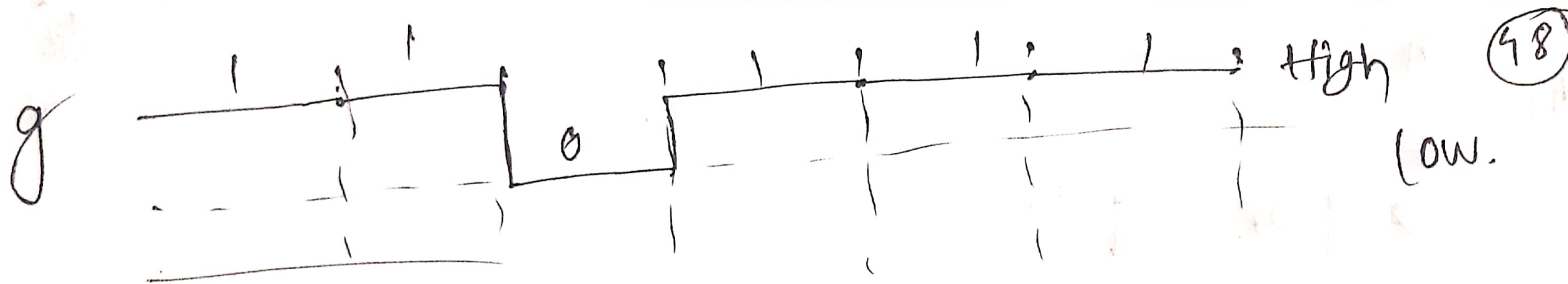


⇒ For 'g':

$$g = A + B + C + D$$

Assume the input signal,





output for g is:

$$g = 110111$$

0